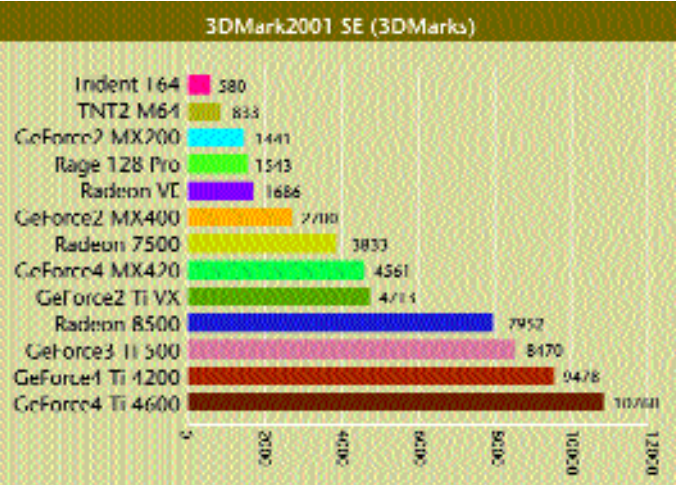




enough data to our GPUs to let them do their thing.

3DMark2001 SE (Direct 3D)

As we move on to the 3DMark2001 SE benchmark, we need to change a few rules. Since this benchmark evaluates much more than just the frame rate, our magic number of 60 fps is no longer valid. The benchmark awards a number, known as a 3DMark to each chipset, after it has performed the 21 tests. Based on the number we can safely comment on the game worthiness of the particular chipset in question.

A synthetic test like this will give you a fair representation of how any Direct3D game will perform on your system. If you want to play the next generation of Direct3D games, like *Unreal Tournament 2003*, then scores upwards of 5000 3DMarks are recommended. Few older generation chipsets can manage to cross that number. A more forgiving game like *Need for Speed 5: Porsche Edition* will require a minimum of 2000 3DMarks to run successfully.

The low-end chipsets, not surprisingly, fare pretty badly. The Trident T64 comes across as the worst of the performers with a lowly 580 3DMarks, closely followed by the TNT2 M64 with a score of 833 3DMarks. The GeForce2 MX200 manages to cross the 1000 barrier with a so-so score of 1441 3DMarks, a performance closely mirrored by the ATi Rage 128 Pro (1543 3DMarks) and the ATi Radeon VE (1686 3DMarks).

The nVidia GeForce2 MX400 posts a very reasonable 2700 3DMarks; a card based on this chipset may not shine under future games, but will be more than happy to take on any current generation game. Even other chipsets like the ATi Radeon 7500; the GeForce4 MX420 and the

5000 3DMark barrier is well and truly broken. The ATi Radeon 8500 posts a healthy score of 7952 3DMarks and is a pretty good choice given its price. Although the GeForce3 Ti 500 scores more than the ATi Radeon 8500 (8470 3DMarks), when you compare it against its newer GeForce4 based cousins, the chipset doesn't look as attractive. The GeForce4 Ti 4200 scores an amazing 9478 points and the GeForce4 Ti 4600 is the only chipset in our roundup that crosses the 10,000 point hurdle to post a phenomenal 10760 3DMarks. This is the best chipset to be sitting in your graphics card, if price is not a factor.

2D performance

Do you click on the Windows Start button and then twiddle your thumbs while the menu appears? Does scrolling through text result in a horrendous lag? If you can identify with any of these situations then you have your video card to blame.

PCMark2002 Pro

It became evident that all of the chipsets in our test can provide decent 2D GUI speed. Even under extreme circumstances when there were multiple windows open, these chipsets did not stutter or slow down the GUI. Quite surprisingly,

GeForce2 Ti VX fail to cross the magic 5000 number with scores of 3833, 4561 and 4713 3DMarks respectively. The GeForce2 Ti VX takes the performance lead against the MX420, once again bringing to light the true nature of the GeForce4 MX420 chipset.

Moving on to the GPUs, our

it was the ATi Rage 128 Pro with a score of 13706.2 lines per second and 1477.1 curves per second that took the lead here. The rest weren't beaten by a mile but lost nonetheless.

The chipset that could render the maximum number of images in a second was the GeForce2 Ti VX with an impressive 639.1 images per second and the GeForce4 MX 420 followed closely with 633.2 images per second.

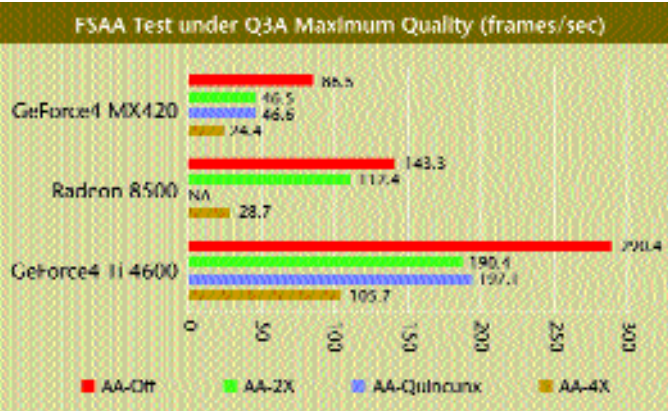
These tests confirm that all modern chipsets can handle 2D content with relative ease and if you are not the least bit inclined towards gaming, then you can pick almost any card and get more than adequate performance for your day-to-day needs.

Image quality

Image quality is a very important factor while evaluating a video chipset. A good frame rate is pretty much useless if the game looks sub-par thanks to textures that show artefacts or pixels that pop in and out of the screen. Similarly, while working with 2D, a good graphics chipset should accurately recreate colours and ensure decent rendering of an operating system's GUI.

Word Document (2D quality)

Windows XP has a nifty feature called Clear Type font, which basically anti-aliases the on-screen text to give a crisp and clear look that is much easier on the eyes. Some of the latest graphics chipsets (like the ATi Radeon 8500 and the nVidia



Given the excellent visual quality that FSAA delivers, a suitable implementation that limits the performance hit when using this feature is a major selling point of all of the new generation chipsets. As this graph demonstrates, at the medium FSAA setting of 2x, both the ATi Radeon 8500 and the Geforce4 Ti 4600 excel at delivering good frame rates. However, the Ti 4600 demonstrates its true power by offering a mind-blowing 105.7 fps at the maximum AA setting of 4x which leaves the 8500 far behind.